4.2
1

a)
$$\begin{array}{r} 231 \\ -128 \\ \hline 103 \\ -64 \\ \hline 39 \\ -32 \\ \hline 7 \\ -4 \\ \hline 3 \\ -2 \\ \hline 1 \\ -1 \\ \hline 0 \end{array}$$

$\boxed{1110\ 0111}$

b)
$$\begin{array}{r} 4532 \\ -4096 \\ \hline 436 \\ -256 \\ \hline 180 \\ -128 \\ \hline 52 \\ -32 \\ \hline 20 \\ -16 \\ \hline 4 \\ -4 \\ \hline 0 \end{array}$$

$\boxed{1\ 0001\ 1011\ 0100}$

c)
$$\begin{array}{r} 97644 \\ -65536 \\ \hline 32108 \\ -16384 \\ \hline 15724 \\ -8192 \\ \hline 7532 \\ -4096 \\ \hline 3436 \\ -2048 \\ \hline 1388 \\ -1024 \\ \hline 364 \\ -256 \\ \hline 108 \\ -64 \\ \hline 44 \\ -32 \\ \hline 12 \\ -8 \\ \hline 4 \\ -4 \\ \hline 0 \end{array}$$

$\boxed{1\ 0111\ 1101\ 0110\ 1100}$

4.23

a) $1\ 1111_2 = 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 16 + 8 + 4 + 2 + 1 = \boxed{31}$

b) $10\ 0000\ 0001_2 = 2^9 + 2^0 = 512 + 1 = \boxed{513}$

c) $1\ 0101\ 0101_2 = 2^8 + 2^6 + 2^4 + 2^2 + 2^0 = 256 + 64 + 16 + 4 + 1 = \boxed{341}$

d) $110\ 1001\ 0001\ 0000_2 = 2^{14} + 2^{13} + 2^{11} + 2^8 + 2^4$

$= 16384 + 8192 + 2048 + 256 + 16 = \boxed{26896}$

4.25

a) $572_8 = \boxed{101\ 111\ 010}$

b) $1604_8 = \boxed{001\ 110\ 000\ 100}$

c) $423_8 = \boxed{100\ 010\ 011}$

d) $2417_8 = \boxed{010\ 100\ 001\ 111}$

4.26

a) $011\ 110\ 111 = \boxed{367_8}$

b) $101\ 010\ 101\ 010 = \boxed{5252_8}$

c) $111\ 011\ 101\ 110\ 111 = \boxed{73567_8}$

d) $101\ 010\ 101\ 010\ 101 = \boxed{52525_8}$

4.2 7

a) $80E_{16} = \boxed{1000\ 0000\ 1110}$

b) $135AB_{16} = \boxed{0001\ 0011\ 0101\ 1010\ 1011}$

c) $ABBA_{16} = \boxed{1010\ 1011\ 1011\ 1010}$

d) $DEFACED_{16} = \boxed{1101\ 1110\ 1111\ 1010\ 1100\ 1110\ 1101}$

4.2 8

$BADFACED_{16}$

$= \boxed{1011\ 1010\ 1101\ 1111\ 1010\ 1100\ 1110\ 1101}$

4.2 23a

$$\begin{array}{r} {\scriptstyle 1\,1\ \ \ } \\ 763_8 \\ +\ 147_8 \\ \hline \boxed{1132_8} \end{array}$$

$$\begin{array}{r} {\scriptstyle 5\ \ 2\ \ } \\ 763_8 \\ \times\ 147_8 \\ \hline 6645 \\ 37140 \\ +\ 76300 \\ \hline \boxed{144305_8} \end{array}$$

4.3 1

a) 21, not prime $3 \cdot 7 = 21$

b) 29, prime $2 \nmid 29$, $3 \nmid 29$, $5 \nmid 29$, $7 > \sqrt{29}$

c) 71, prime $2 \nmid 71$, $3 \nmid 71$, $5 \nmid 71$, $7 \nmid 71$, $11 > \sqrt{71}$

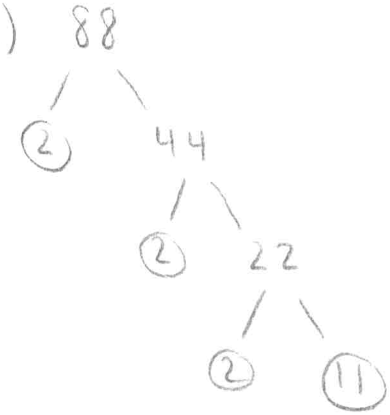
d) 97, prime $2 \nmid 97$, $3 \nmid 97$, $5 \nmid 97$, $7 \nmid 97$, $11 > \sqrt{97}$

e) 111, not prime $3 \cdot 37 = 111$

f) 143, not prime $11 \cdot 13 = 143$

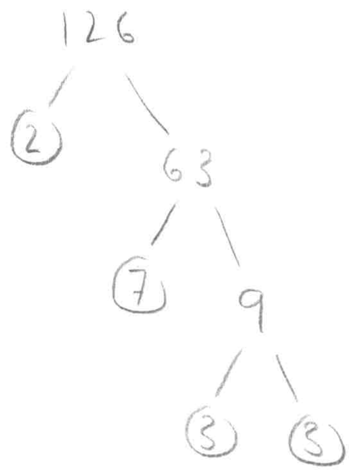
4.3
3

a)



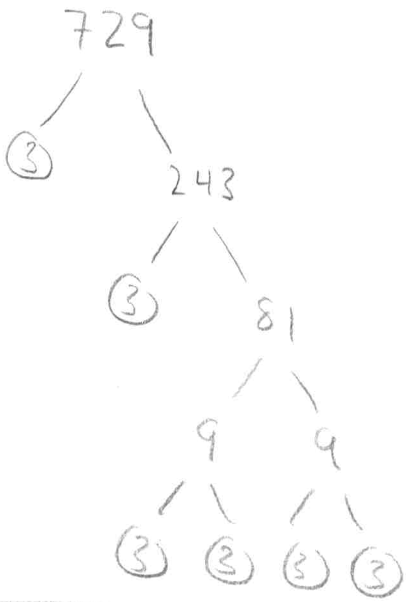
$\boxed{88 = 2^3 \cdot 11}$

b)



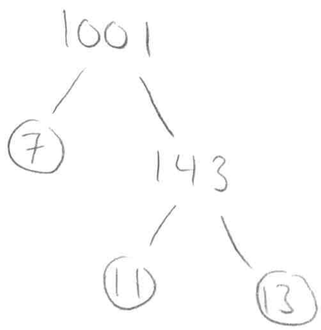
$\boxed{126 = 2 \cdot 3^2 \cdot 7}$

c)



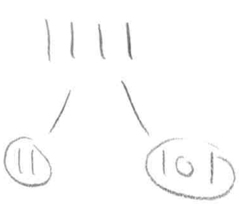
$\boxed{729 = 3^6}$

d)



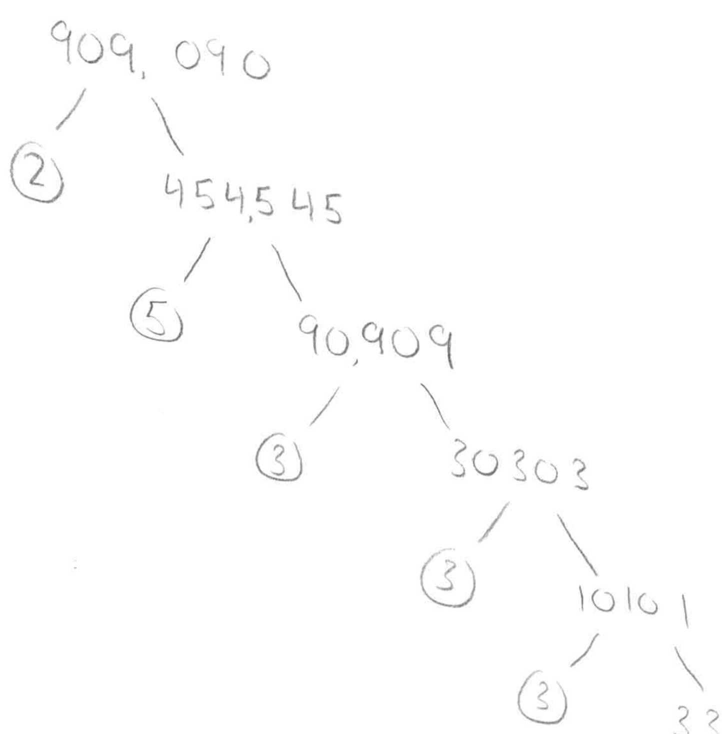
$\boxed{1001 = 7 \cdot 11 \cdot 13}$

e)



$\boxed{1111 = 11 \cdot 101}$

f)



f) 909,090

2   454,545

5   90,909

3   30,303

3   10,101

3   3367

7   481

13   37

$$\boxed{909{,}090 = 2 \cdot 3^3 \cdot 5 \cdot 7 \cdot 13 \cdot 37}$$

4.3 1.4

gcd(1, 12) = 1 ✓

gcd(2, 12) = 2 X

gcd(3, 12) = 3 X

gcd(4, 12) = 4 X

gcd(5, 12) = 1 ✓

gcd(6, 12) = 6 X

gcd(7, 12) = 1 ✓

gcd(8, 12) = 4 X

gcd(9, 12) = 3 X

gcd(10, 12) = 2 X

gcd(11, 12) = 1 ✓

$\boxed{1, 5, 7, 11}$

4.3
16

a) 21, 34, 55

$21 = 3 \cdot 7$ $34 = 2 \cdot 17$ $55 = 5 \cdot 11$

no common prime factors, so $\boxed{\text{pairwise relatively prime}}$

b) 14, 17, 85

$85 = 5 \cdot 17$, shares a common prime factor with 17

$\boxed{\text{not pairwise relatively prime}}$

c) 25, 41, 49, 64

$25 = 5^2$, 41 is prime, $49 = 7^2$, $64 = 8^2$

no common prime factors, so $\boxed{\text{pairwise relatively prime.}}$

d) 17, 18, 19, 23

17 is prime, $18 = 2 \cdot 3^2$, 19 is prime, 23 is prime

no common prime factors, so $\boxed{\text{pairwise relatively prime}}$

4.3
24

a) $2^2 \cdot 3^3 \cdot 5^2$

b) $2 \cdot 3 \cdot 11$

c) 17

d) 1

e) 5

f) $2 \cdot 3 \cdot 5 \cdot 7$

4.3 27

a) $2^{11} \cdot 3^7 \cdot 5^9 \cdot 7^3$

b) $2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17$

c) 23^{31}

d) $41 \cdot 43 \cdot 53$

e) $2^{12} \cdot 3^{13} \cdot 5^{17} \cdot 7^{21}$

f) does not exist because every multiple of 0 is 0.

4.3 33

a) $\gcd(12, 18)$

$= \gcd(18 \bmod 12, 12) = \gcd(6, 12)$

$= \gcd(12 \bmod 6, 6) = \gcd(0, 6)$

$= \boxed{6}$

b) $\gcd(111, 201)$

$= \gcd(201 \bmod 111, 111) = \gcd(90, 111)$

$= \gcd(111 \bmod 90, 90) = \gcd(21, 90)$

$= \gcd(90 \bmod 21, 21) = \gcd(6, 21)$

$= \gcd(21 \bmod 6, 6) = \gcd(3, 6)$

$= \gcd(6 \bmod 3, 3) = \gcd(0, 3)$

$= \boxed{3}$

c) $\gcd(1001, 1331)$

$= \gcd(1331 \bmod 1001, 1001) = \gcd(330, 1001)$

$= \gcd(1001 \bmod 330, 330) = \gcd(11, 330)$

$= \gcd(330 \bmod 11, 11) = \gcd(0, 11)$

$= \boxed{11}$

d) $\gcd(12345, 54321)$

$= \gcd(54321 \bmod 12345, 12345) = \gcd(4941, 12345)$

$= \gcd(12345 \bmod 4941, 4941) = \gcd(2463, 4941)$

$= \gcd(4941 \bmod 2463, 2463) = \gcd(15, 2463)$

$= \gcd(2463 \bmod 15, 15) = \gcd(3, 15)$

$= \gcd(15 \bmod 3, 3) = \gcd(0, 3)$

$= \boxed{3}$

e) $\gcd(1000, 5040)$

$= \gcd(5040 \bmod 1000, 1000) = \gcd(40, 1000)$

$= \gcd(1000 \bmod 40, 40) = \gcd(0, 40)$

$= \boxed{40}$

f) $\gcd(9888, 6060)$

$= \gcd(9888 \bmod 6060, 6060) = \gcd(3828, 6060)$

$= \gcd(6060 \bmod 3828, 3828) = \gcd(2232, 3828)$

$= \gcd(3828 \bmod 2232, 2232) = \gcd(1596, 2232)$

$= \gcd(2232 \bmod 1596, 1596) = \gcd(636, 1596)$

$= \gcd(1596 \bmod 636, 636) = \gcd(324, 636)$

$= \gcd(636 \bmod 324, 324) = \gcd(312, 324)$

$= \gcd(324 \bmod 312, 312) = \gcd(12, 312)$

$= \gcd(312 \bmod 12, 12) = \gcd(0, 12)$

$= \boxed{12}$

4.3
34

$\gcd(21, 34)$

$= \gcd(34 \bmod 21, 21) = \gcd(13, 21)$

$= \gcd(21 \bmod 13, 13) = \gcd(8, 13)$

$= \gcd(13 \bmod 8, 8) = \gcd(5, 8)$

$= \gcd(8 \bmod 5, 5) = \gcd(3, 5)$

$= \gcd(5 \bmod 3, 3) = \gcd(2, 3)$

$= \gcd(3 \bmod 2, 2) = \gcd(1, 2)$

$= \gcd(2 \bmod 1, 1) = \gcd(0, 1)$

$= \boxed{1}$ $\boxed{7 \text{ divisions}}$

4.3
39h

$\gcd(3454, 4666)$

$= \gcd(4666 \bmod 3454,\ 3454) = \gcd(1212, 3454)$

$= \gcd(3454 \bmod 1212,\ 1212) = \gcd(1030, 1212)$

$= \gcd(1212 \bmod 1030,\ 1030) = \gcd(182, 1030)$

$= \gcd(1030 \bmod 182,\ 182) = \gcd(120, 182)$

$= \gcd(182 \bmod 120,\ 120) = \gcd(62, 120)$

$= \gcd(120 \bmod 62,\ 62) = \gcd(58, 62)$

$= \gcd(62 \bmod 58,\ 58) = \gcd(4, 58)$

$= \gcd(58 \bmod 4,\ 4) = \gcd(2, 4)$

$= \gcd(4 \bmod 2,\ 2) = \gcd(0, 2)$

$= 2$

$2 = 58 - 14 \cdot 4$

$= 58 - 14 \cdot (62 - 58) = 15 \cdot 58 - 14 \cdot 62$

$= 15 \cdot (120 - 62) - 14 \cdot 62 = 15 \cdot 120 - 29 \cdot 62$

$= 15 \cdot 120 - 29 \cdot (182 - 120) = 44 \cdot 120 - 29 \cdot 182$

$= 44 \cdot (1030 - 5 \cdot 182) - 29 \cdot 182 = 44 \cdot 1030 - 249 \cdot 182$

$= 44 \cdot 1030 - 249 (1212 - 1030) = 293 \cdot 1030 - 249 \cdot 1212$

$= 293 \cdot (3454 - 2 \cdot 1212) - 249 \cdot 1212 = 293 \cdot 3454 - 835 \cdot 1212$

$= 293 \cdot 3454 - 835 (4666 - 3454) = \boxed{1128 \cdot 3454 - 835 \cdot 4666}$

4.3
42

$\gcd(356, 252)$

$= \gcd(356 \bmod 252, 252) = \gcd(104, 252)$ $\quad q_1 = 1 \quad r_2 = 104$

$= \gcd(252 \bmod 104, 104) = \gcd(44, 104)$ $\quad q_2 = 2 \quad r_3 = 44$

$= \gcd(104 \bmod 44, 44) = \gcd(16, 44)$ $\quad q_3 = 2 \quad r_4 = 16$

$= \gcd(44 \bmod 16, 16) = \gcd(12, 16)$ $\quad q_4 = 2 \quad r_5 = 12$

$= \gcd(16 \bmod 12, 12) = \gcd(4, 12)$ $\quad q_5 = 1 \quad r_6 = 4$

$= \gcd(12 \bmod 4, 4) = \gcd(0, 4)$ $\quad q_6 = 3$

$= 4$

$S_2 = S_0 - q_1 S_1 = 1 - 1 \cdot 0 = 1$ $\qquad t_2 = t_0 - q_1 t_1 = 0 - 1 \cdot 1 = -1$

$S_3 = S_1 - q_2 S_2 = 0 - 2 \cdot 1 = -2$ $\qquad t_3 = t_1 - q_2 t_2 = 1 - 2 \cdot (-1) = 3$

$S_4 = S_2 - q_3 S_3 = 1 - 2 \cdot (-2) = 5$ $\qquad t_4 = t_2 - q_3 t_3 = -1 - 2 \cdot 3 = -7$

$S_5 = S_3 - q_4 S_4 = -2 - 2 \cdot 5 = -12$ $\qquad t_5 = t_3 - q_4 t_4 = 3 - 2 \cdot (-7) = 17$

$S_6 = S_4 - q_5 S_5 = 5 - 1 \cdot (-12) = 17$ $\qquad t_6 = t_4 - q_5 t_5 = -7 - 1(17) = -24$

$$\boxed{4 = 17 \cdot 356 - 24 \cdot 252}$$